Week 08: October 19-23
Communities, Biomes and Ecosystems

Subject/:الموضوع Biology **Unit Title: عنوان الوحدة Ecology** **Gradeالصف : 9**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1:	**SUMMATIVE TEST** I can demonstrate my understanding of the concepts we have learned by completing the Criterion A Summative Assessment: • MUST — Levels 1–2: I can recall and identify key biological facts, terms, structures, and processes with some accuracy. • SHOULD — Levels 3–4: I can describe and explain biological concepts and use relevant evidence or examples to show my understanding. • SHOULD — Levels 5–6: I can apply and analyze my biological knowledge to unfamiliar situations, interpret information, and explain relationships using appropriate scientific terminology. • COULD — Levels 7–8: I can apply, analyze, and evaluate biological concepts accurately and independently, making well-supported conclusions and demonstrating a sophisticated understanding of unfamiliar problems.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Study guide to be shared n Toddle and Teams

Lesson 2	I can label freshwater ecosystem zones and compare oligotrophic and eutrophic conditions by applying my understanding to propose a water-conservation strategy for a freshwater ecosystem.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Aquatic Ecosystems.pdf Reading Essentials- Terrestrial Biomes.pdf Reading Essentials- Community Ecology.pdf
Lesson 3:	I can label marine and transitional ecosystem zones and compare mangrove and open-ocean conditions by applying my understanding to design a marine-conservation campaign for a coastal site.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Aquatic Ecosystems.pdf Reading Essentials- Terrestrial Biomes.pdf Reading Essentials- Community Ecology.pdf
Lesson 4	I can identify the world's major biomes and compare the abiotic factors of two biomes by applying my understanding to design an educational exhibit connecting biome science to real-world adaptations.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Aquatic Ecosystems.pdf Reading Essentials- Terrestrial Biomes.pdf Reading Essentials- Community Ecology.pdf
Homework الواجبات			

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	

Week # 08 الأسبوع#

Technology Integration دمج التكنولوجيا	Short video clip on a real land-restoration project Virtual field-trip video clip of a protected freshwater site (e.g. Wadi Wurayah, UAE) Google Earth virtual tour of a mangrove reserve and nearby coastline Short documentary clip on Gulf/reef coral bleaching World biome map (Google Earth)
Cross-curricular links الروابط بين المواد الدراسية	Lesson 2 Geography — the water cycle and regional hydrology Mathematics — percentage and ratio calculations of freshwater distribution History/Heritage — traditional irrigation systems, e.g. the falaj system of Al Ain (UNESCO World Heritage) Environmental Science — national and global water-security strategies Lesson 3 Geography — coastal geomorphology and tidal patterns History/Heritage — the Gulf region's pearl-diving tradition and dhow-building Economics — the fishing industry and coastal tourism/development Art & Design — campaign poster design for the DOK 4 task Lesson 4 Geography — global climate zones and biome mapping History/Heritage — Bedouin desert life, camel herding, date-palm cultivation, majlis culture Physics/Chemistry — heat and water-conservation adaptations of desert organisms
External Assessment Integration دمج التقييمات الخارجية	N/A
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking — comparing, contrasting and evaluating succession pathways

	ATL Communication — structuring a clear written comparison paragraph ATL Self-management — using success-criteria checklists to monitor progress ATL Research — gathering and citing real freshwater-scarcity data ATL Self-management — pacing a data-calculation task within a time limit ATL Social skills — small-group case-study discussion and consensus-building
Standards **المعايير**	**HS-LS2-6: Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.** **Build-to:** **HS-LS2-2, HS-LS2-5, HS-LS2-7**
Inquiry/ TOK Question **البحث والاستقصاء أسئلة TOK**	Can a damaged ecosystem ever truly return to what it was before, or does change always leave something behind? *Is freshwater a right or a resource — and does an ecosystem's value depend on how scarce it is?*
Success Criteria معايير النجاح (Measurable)	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Lesson 2:** MUST: correctly label the headwater through mouth on a river diagram and all three lake zones, and state the percentage of Earth's water that is freshwater. SHOULD: compare at least two factors (e.g., flow speed, oxygen, sediment, or nutrients) and accurately explain how they influence differences in species diversity using correct zone terminology. COULD: develop a proposal that identifies a real freshwater feature, cites at least one relevant statistic, proposes two concrete conservation actions, and justifies its ecological and cultural importance. **Lesson 3:** MUST: correctly label at least 7 of the 8 named zones/terms across the intertidal and open-ocean diagrams. SHOULD: reference at least 3 abiotic factors (salinity, tide, and light) and correctly contrast at least 2 organism adaptations between mangrove and open-ocean environments. COULD: develop a campaign that names a real coastal site, identifies at least 2 threats, proposes at least 2 conservation actions, and weighs conservation needs against development or heritage needs. **Lesson 4:**

	MUST: correctly match at least 8 of the 9 biomes to their climate and location descriptions. SHOULD: compare at least 3 abiotic factors between two biomes and connect each factor to at least 2 specific organism adaptations. COULD: develop an exhibit plan that includes at least 3 species/adaptations from a chosen biome, at least 1 connection to human heritage or traditional knowledge, and a clear educational message.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Choose an item. Global Understanding التفاهم العالمي Choose an item. المجال **Area:** environmental stewardship, conservation, and sustainable development as shared responsibilities التطبيق Implementation: Students examine how the UAE's greening, desert afforestation, and land-reclamation efforts transformed previously arid landscapes. They connect **ecological succession**—pioneer species, soil development, increasing biodiversity, and community change—to UAE examples such as **Al Ain's greening initiatives**. Students then design a restoration proposal for a degraded UAE habitat, explaining how ecological succession can support sustainable environmental development. Students investigate the traditional **falaj irrigation systems of Al Ain** and the conservation of freshwater ecosystems in **Wadi Wurayah**. They connect these UAE examples to the ecological concepts of **water scarcity, zonation, resource availability, and ecosystem management**. Students evaluate how careful water management enables communities and ecosystems to persist in an arid environment and propose strategies for sustainable freshwater use.